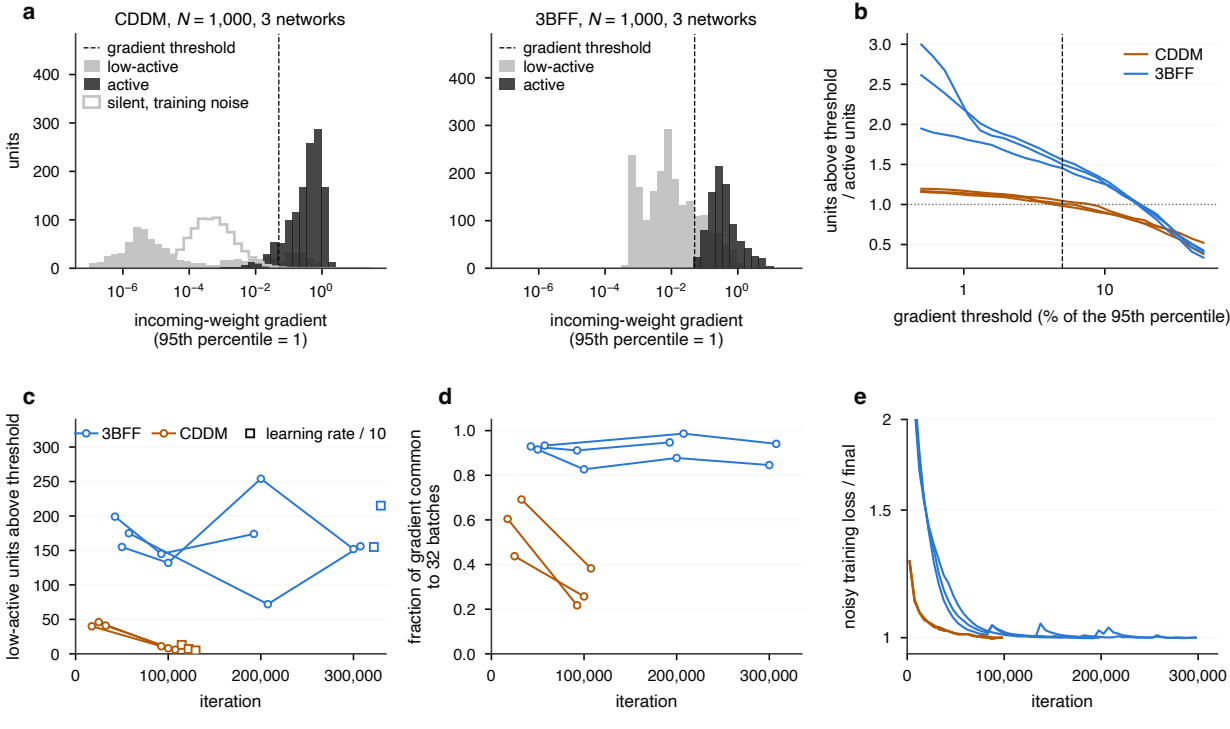
a
CDDM, N = 1,000, 3 networks
gradient threshold
low-active
active
silent, training noise
units
400
300
200
100
0
10^−6
10^−4
10^−2
10^0
incoming-weight gradient
(95th percentile = 1)
3BFF, N = 1,000, 3 networks
gradient threshold
low-active
active
400
300
200
100
0
10^−6
10^−4
10^−2
10^0
incoming-weight gradient
(95th percentile = 1)
b
CDDM
3BFF
units above threshold
/ active units
3.0
2.5
2.0
1.5
1.0
0.5
1
10
gradient threshold (% of the 95th percentile)
c
3BFF
CDDM
learning rate / 10
low-active units above threshold
300
250
200
150
100
50
0
0
100,000
200,000
300,000
iteration
d
fraction of gradient common
to 32 batches
1.0
0.8
0.6
0.4
0.2
0.0
0
100,000
200,000
300,000
iteration
e
noisy training loss / final
2
1.5
1
0
100,000
200,000
300,000
iteration